Adit Shah

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Education

University of California, Berkeley — Berkeley, CA

B.S. Electrical Engineering & Computer Sciences (GPA: 3.98) — *Aug. 2021 – May 2025*

Selected Coursework: Machine Learning, AI Meets Bio & Chem, Algorithms for Computational Biology, Computational Genomics, Convex Optimization, Probability Theory & Random Processes, Statistical Mechanics, Classical Mechanics, Electricity & Magnetism, Discrete Mathematics, Computer Architecture, Artificial Intelligence, Operating Systems

Experience

Columbia University — New York, NY

Research Associate – AlQuraishi Lab — *Aug. 2026 – Present*

- Developing model adaptation techniques to steer AlphaFold3 with experimental data and extend accurate structure determination to flexible proteins, RNA, and complexes
- Training encoders tailored to each type of experimental data to feed a shared conditioning framework
- Recovering up to 77% of idle GPU capacity on Columbia's shared cluster by building a simulator of past usage to test scheduling policies and collaborating with PIs and Columbia IT on implementation
- Building a foundation in biophysics by reading *Physical Biology of the Cell* and participating in the lab's focus group on conformations, including presenting literature on generative models for sampling transition pathways

Until Labs — South San Francisco, CA

Research Engineer — *Jun. 2024 – Jul. 2026*

- First computational hire; built the company's scientific data platform and analysis pipelines with scientists across teams
- Developed machine learning models and simulation pipelines to discover new cryoprotectants

Broad Institute of MIT and Harvard — Cambridge, MA

Research Intern – Imaging Platform — *Jan. 2024 – May 2024*

- Found a flaw in drug discovery benchmarks where many assays were trivially predicted by cell count rather than finer morphological features, leading to a co-authored publication in *Nature Communications*
- Built modular evaluation framework comparing cellular imaging, chemical structure, and gene expression for predicting biochemical assays, supporting flexible combinations of embeddings and models
- Established SkyPilot cloud infrastructure and GPU resource pipelines adopted by the broader lab

Berkeley Artificial Intelligence Research (BAIR) — Berkeley, CA

Undergraduate Researcher – Berkeley Speech Group — *Aug. 2023 – Feb. 2024*

- Curated and preprocessed datasets across three silent-speech recording modalities to demonstrate generalization of streaming neural speech decoding architectures
- Co-authored a *Nature Neuroscience* paper from an active clinical trial restoring speech to paralyzed individuals

Undergraduate Researcher – CIRCLES Consortium — *Dec. 2021 – May 2023*

- Trained and evaluated RL controllers for autonomous vehicles to reduce highway traffic congestion, including observation space design, reward shaping, and training curricula
- Demonstrated >15% improvement in fuel efficiency in simulation with just 4% autonomous vehicle penetration, leading to a co-first-authored publication at IEEE ITSC
- Deployed algorithm on 100+ vehicles for a week-long field test in Nashville, the largest autonomous vehicle traffic experiment to date, resulting in two *IEEE Control Systems* papers

EPFL (École Polytechnique Fédérale de Lausanne) — Lausanne, Switzerland

Summer@EPFL Research Fellow – MLBio Lab — *May 2023 – Aug. 2023*

- Developed a modular framework benchmarking meta-learning approaches to few-shot learning on biomedical tasks including drug response prediction, antibiotic resistance, and protein function annotation

Selected Projects & Initiatives

Machine Learning for Biology | *Course Projects & Presentations* Aug. 2024 – May 2025

- *Fusion Protein Prioritization & Design*: Classified oncogenic fusions from protein language model embeddings at 0.877 AUC, and experimented with steering a LoRA-tuned protein diffusion model via a cross-attention adapter
- *Chromatin State Prediction*: Extended the DeepSEA genomic CNN with dilated convolutions, attention, and longer input sequences, finding that a 2,000 bp context window improved performance
- *Protein Foundation Models*: Presented a three-hour literature review of ESM3 to a graduate course, covering how tokenizing structure with a VQ-VAE enables joint generative modeling of protein sequence, structure, and function

Neurotech @ Berkeley | *Software Division Lead* Aug. 2022 – Jan. 2024

- Led a division of 30 Berkeley students working on projects related to software, ML, and neuroscience
- Elected President for organization of 120 members

CS 61B: Data Structures | *Undergraduate Student Instructor & Tutor* Jun. 2022 – Jun. 2023

- Taught discussion and laboratory sections with 50+ students to teach concepts such as object-oriented programming, asymptotic analysis, hashing, sorting, and graph algorithms
- Hosted office hours, graded student assignments/exams, and developed course curriculum (assignments and projects) for 1,500+ students

Publications

Counting cells can accurately predict small-molecule bioactivity benchmarks. S. Seal, W. Dee, **A. Shah**, ..., S. Singh, O. Spjuth, A. Bender, A.E. Carpenter. *Nature Communications*, 2026.

A streaming brain-to-voice neuroprosthesis to restore naturalistic communication. K.T. Littlejohn, C.J. Cho, J.R. Liu, A.B. Silva, B. Yu, V.R. Anderson, C.M. Kurtz-Miott, S. Brosler, A.P. Kashyap, I.P. Hallinan, **A. Shah**, A. Tu-Chan, K. Ganguly, D.A. Moses, E.F. Chang, G.K. Anumanchipalli. *Nature Neuroscience*, 28, pp. 902–912, 2025.

Traffic control via connected and automated vehicles (CAVs): An open-road field experiment with 100 CAVs. J.W. Lee, H. Wang, K. Jang, N. Lichtlé, A. Hayat, M. Bunting, A. Alanqary, W. Barbour, Z. Fu, X. Gong, G. Gunter, S. Hornstein, A.R. Kreidieh, M.W. Nice, W.A. Richardson, **A. Shah**, E. Vinitsky, F. Wu, ..., A.M. Bayen. *IEEE Control Systems*, 45(1), pp. 28–60, 2025.

Reinforcement learning-based oscillation dampening: Scaling up single-agent reinforcement learning algorithms to a 100-autonomous-vehicle highway field operational test. K. Jang, N. Lichtlé, E. Vinitsky, **A. Shah**, M. Bunting, M. Nice, B. Piccoli, B. Seibold, D.B. Work, M.L. Delle Monache, J. Sprinkle, J.W. Lee, A.M. Bayen. *IEEE Control Systems*, 45(1), pp. 61–94, 2025.

Traffic smoothing controllers for autonomous vehicles using deep reinforcement learning and real-world trajectory data. N. Lichtlé*, K. Jang*, **A. Shah***, E. Vinitsky*, J.W. Lee, A.M. Bayen. *2023 IEEE International Conference on Intelligent Transportation Systems (ITSC)*, pp. 4346–4351, 2023.

** Equal contribution*

Awards & Honors

High Honors in Electrical Engineering and Computer Sciences | *UC Berkeley* 2025

Regents' & Chancellor's Scholar | *UC Berkeley* 2021 – 2025

Summer@EPFL Research Fellowship | *EPFL* 2023

Technical Skills

Languages: Python, Java, C/C++, Rust, Kotlin, JavaScript
ML Frameworks: PyTorch, Scikit-Learn, XGBoost, OpenCV, Stable Baselines, OpenAI Gym
ML Methods: CNNs, Reinforcement Learning (PPO), Few-Shot Learning, VQ-VAE
Computational Science: OpenFold3, Protein Language Models (ESM), RDKit, ORCA, PyMOL
Data: NumPy, SciPy, Pandas, Polars, DuckDB, PostgreSQL, SQLAlchemy, Marimo
Infrastructure & Tools: AWS (SQS, S3, EC2 Spot), Azure, Modal, SkyPilot, Ray, SLURM, Docker, Git, gRPC